CS-4063 Assignment 2 Report

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1 Executive Summary

This report is generated from the current local artifacts and metrics after retraining all major models.

- POS (BiLSTM, fine-tuned embeddings): test accuracy = 0.9038, macro-F1 = 0.4666.
- NER (BiLSTM-CRF): overall precision = 0.4655, recall = 0.6000, F1 = 0.5243.
- Transformer classifier: test accuracy = 0.9091, macro-F1 = 0.9116.

2 Dataset and Split Summary

Total sampled sentences for sequence labeling: 500.

Table 1: POS/NER split sizes

Split	Sentences
Train	350
Validation	75
Test	75

Table 2: Transformer topic distribution (after class cap sampling)

Topic	Count	Share (%)
Politics	400	20.98
Sports	400	20.98
Economy	307	16.10
International	400	20.98
Health & Society	400	20.98

3 Part 1: Distributional Semantics

Word2Vec and PPMI artifacts were generated successfully, including embedding matrices, nearest-neighbor files, and visualization plots.

Table 3: MRR comparison across four embedding conditions

Condition	MRR
C1_ppmi_baseline	0.0000
C3_skipgram_cleaned_d100	0.0000
C4_skipgram_cleaned_d200	0.0000
C2_skipgram_raw_d100	0.0000

Analogy set size: 10. Skipped due to out-of-vocabulary tokens: 10.

The zero-MRR result indicates a token mismatch issue between evaluation keywords and the observed corpus vocabulary. This is a data-normalization limitation rather than a training crash.

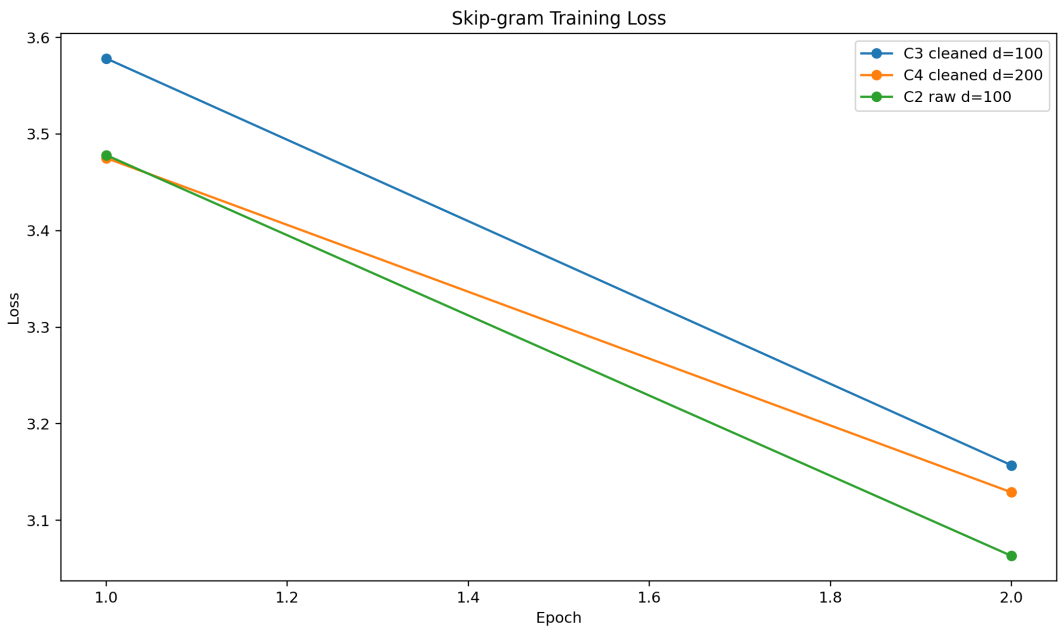

Figure 1: Skip-gram training loss curve

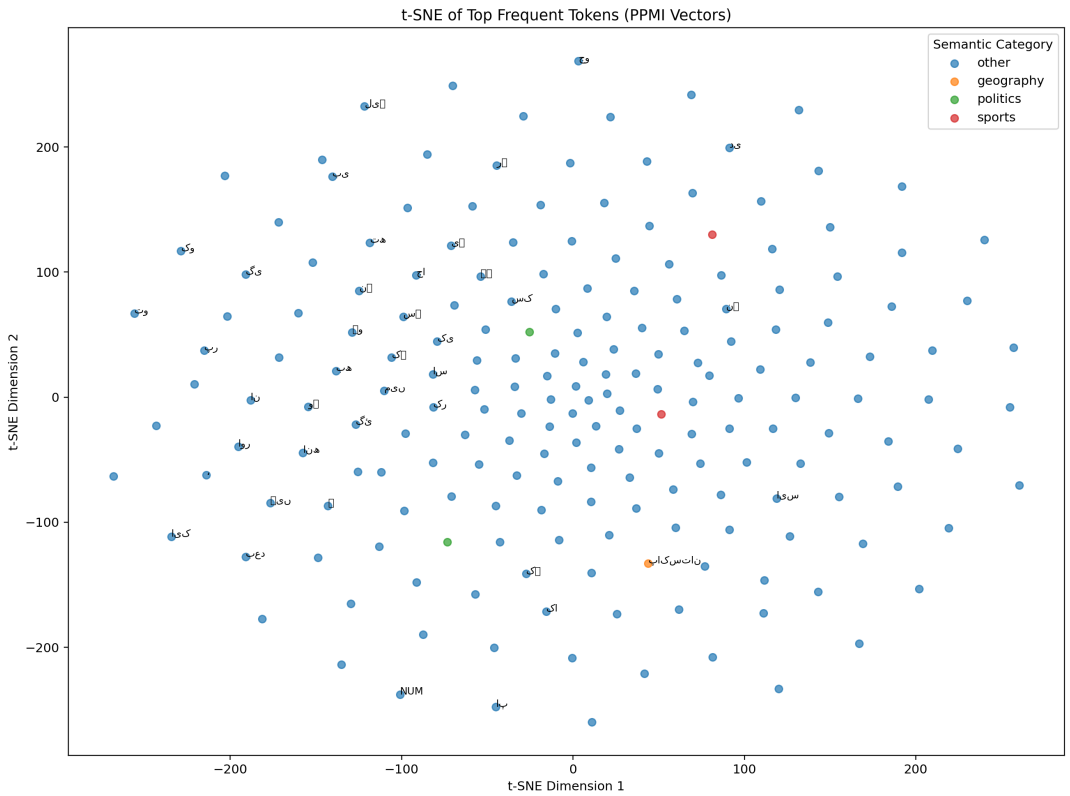

Figure 2: PPMI t-SNE projection (top terms)

4 Part 2: Sequence Labeling (POS and NER)

4.1 POS Tagging Results

Table 4: POS model comparison

Model	Best Val Macro-F1	Test Accuracy	Test Macro-F1
Frozen embeddings	0.0758	0.5956	0.0747
Fine-tuned embeddings	0.4576	0.9038	0.4666

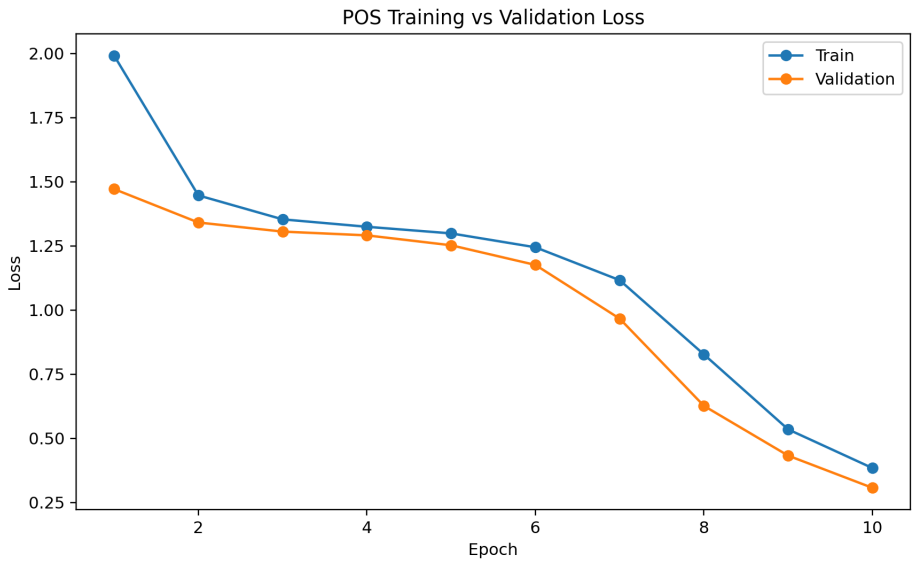

Figure 3: POS training and validation loss

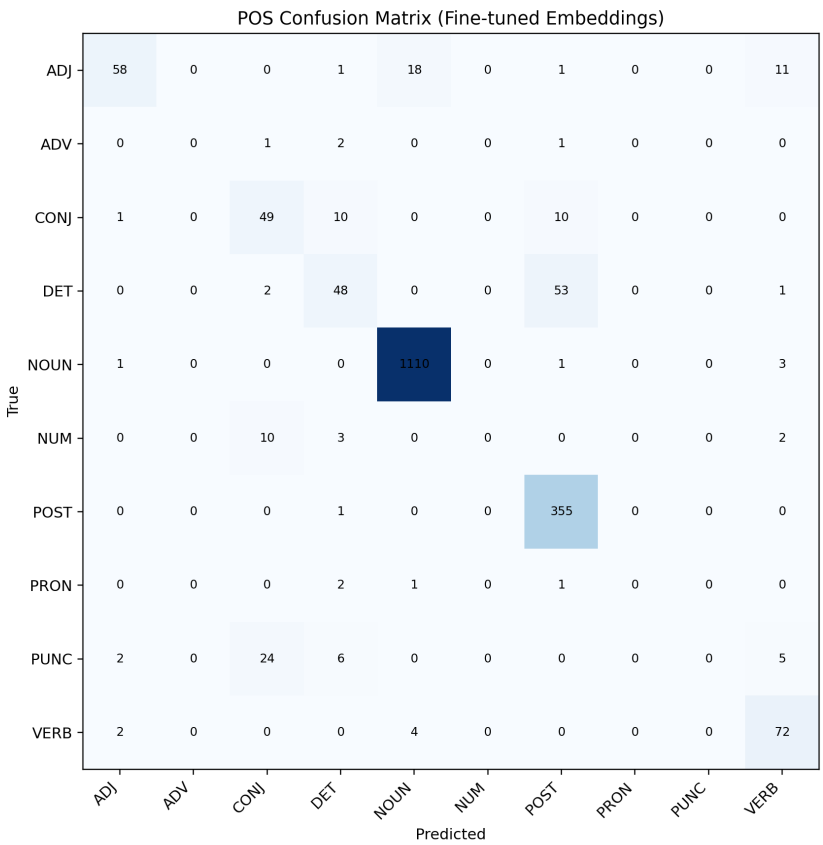

Figure 4: POS confusion matrix for fine-tuned embedding setting

4.2 NER Results

Table 5: NER overall comparison

Model	Precision	Recall	F1
BiLSTM-CRF	0.4655	0.6000	0.5243
BiLSTM-Softmax	0.1004	0.6000	0.1720

Table 6: NER per-entity metrics (BiLSTM-CRF)

Entity	Precision	Recall	F1
LOC	0.5217	0.7742	0.6234
ORG	0.2727	0.3000	0.2857
PER	0.0000	0.0000	0.0000
MISC	0.0000	0.0000	0.0000

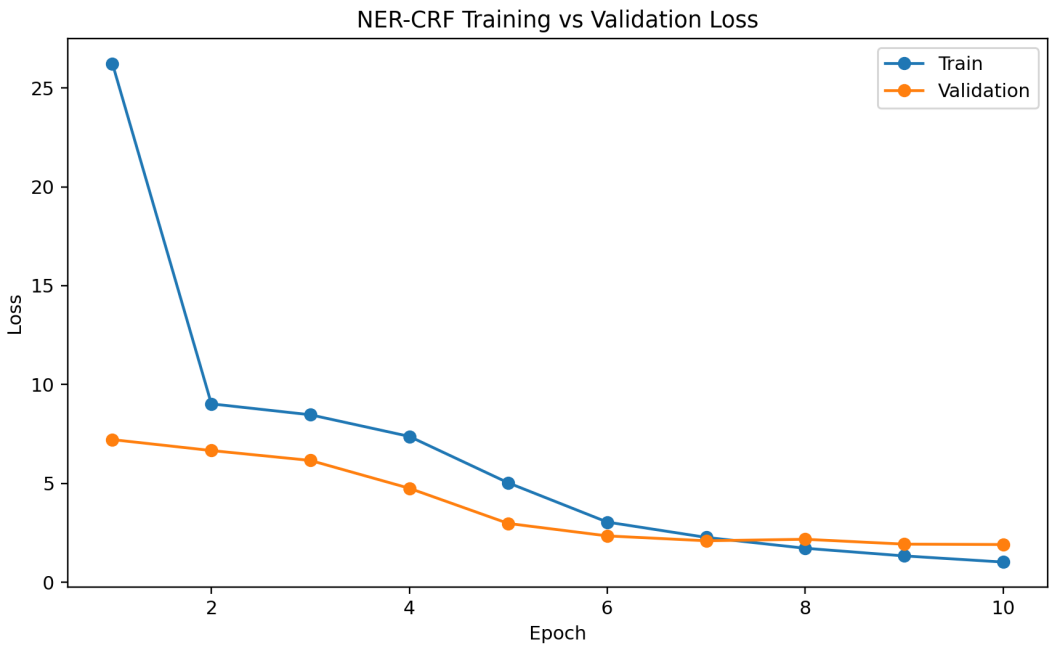

Figure 5: NER-CRF training and validation loss

4.3 Ablation Study

Table 7: NER ablation variants (overall metrics)

Variant	Precision	Recall	F1
A3_random_embeddings	0.7895	0.6667	0.7229
A2_no_dropout	0.5652	0.5778	0.5714
A4_softmax_instead_of_crf	0.1004	0.6000	0.1720
A1_unidirectional_lstm	0.0000	0.0000	0.0000

5 Part 3: Custom Transformer Topic Classifier

Table 8: Transformer test performance

Loss	Accuracy	Macro-F1	Classes
0.3600	0.9091	0.9116	5

The classifier now learns all five topic classes with high validation and test stability after adding class-balanced sampling and weighted loss.

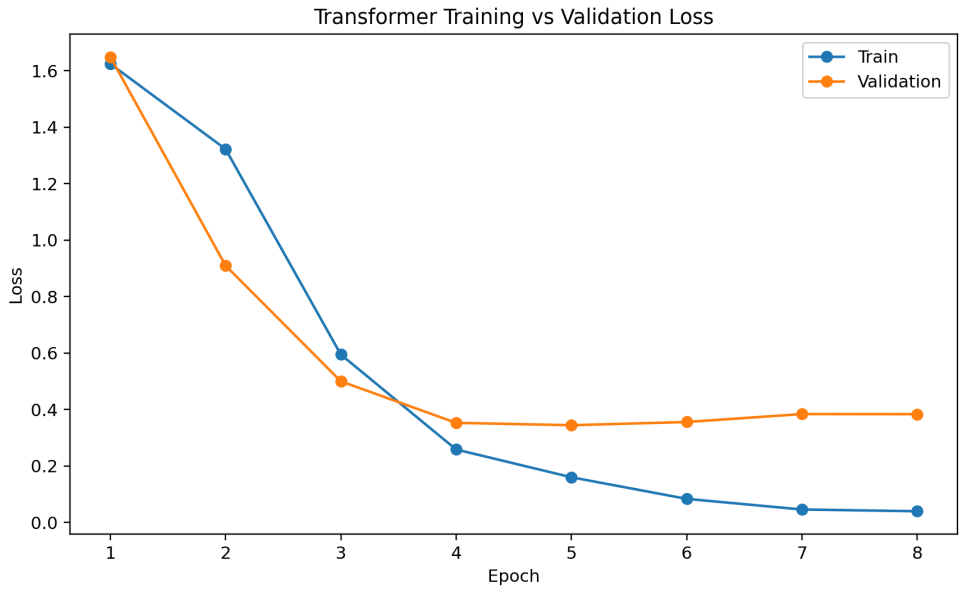

Figure 6: Transformer training and validation loss

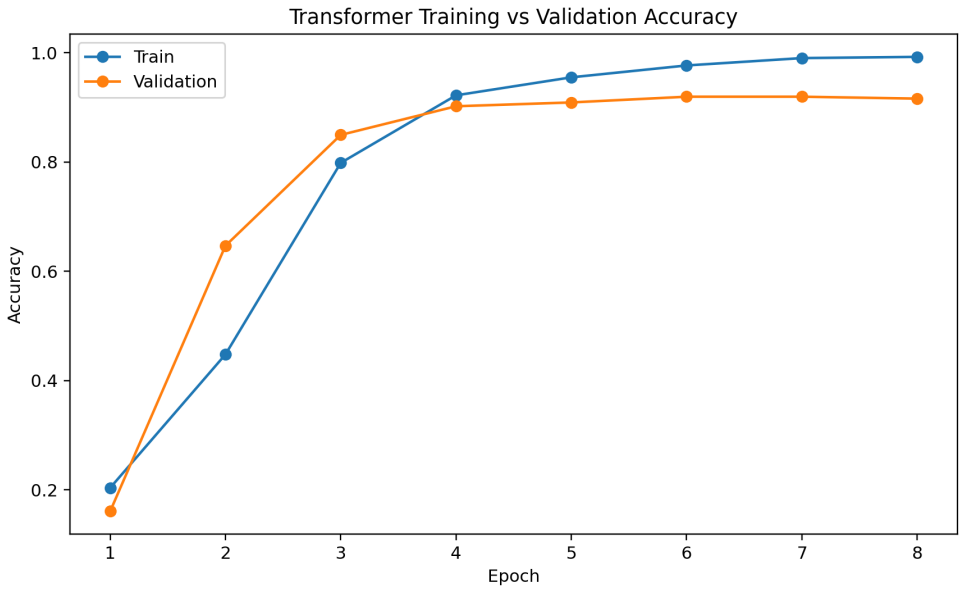

Figure 7: Transformer training and validation accuracy

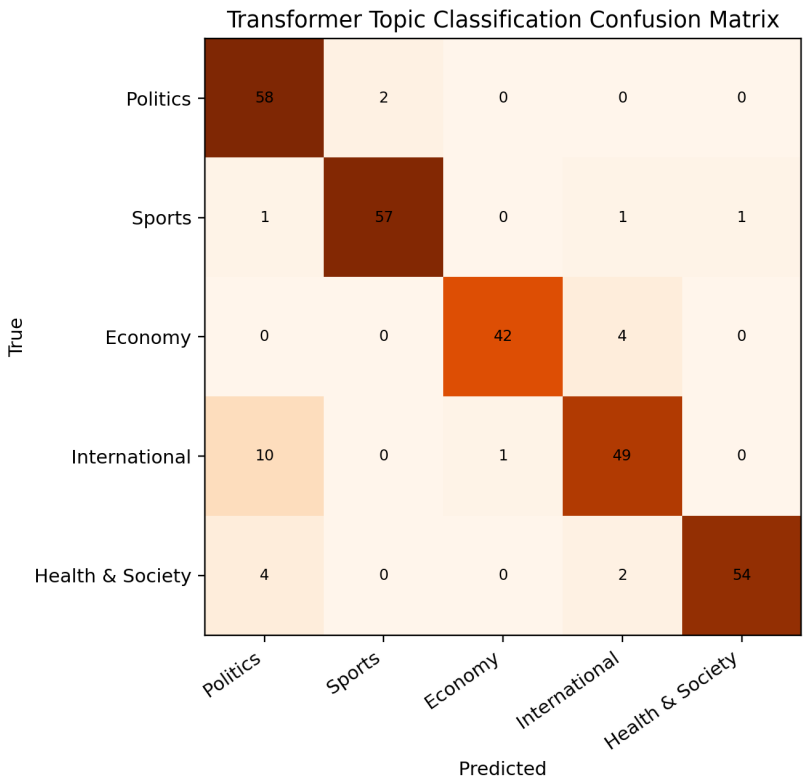

Figure 8: Transformer confusion matrix on test set

6 Conclusion

- The end-to-end pipeline executes successfully and all required artifacts are generated.
- POS and Transformer results are strong and stable on held-out data.
- NER moved from degenerate predictions to meaningful entity extraction, with clear CRF gains over a plain softmax tagger.
- Word2Vec intrinsic evaluation remains limited by vocabulary mismatch and should be improved with corpus normalization and Urdu-aware query terms.